Where should a small controller's skill live?

Routines in a library, route choice in a table of verified outcomes

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In one sentence

Moving the decomposition routine out of the weights into a prebound library, and the direct-versus-decompose choice into a table of verified outcomes (retrieval keyed by the input, as in RAG), lets an *untrained* 4B controller match or beat every controller we trained, at lower cost, on both task families.

Where we were: learned selective recursion, and routing collapse.

Setup. Qwen3-4B controls a recursive language model (RLM): it writes Python cells, calls helper models on chunks, and submits an answer. Two families: a synthetic store log (8 to 512 lines, five question forms) and OOLONG-synth (1k to 32k tokens).

What worked. Teaching by replaying a scripted expert, then RL with a verifiable reward: held-out OOLONG 0.567 (taught) to 0.652 (RL); direct answering 0.208.

What did not. Every RL form lost the *route choice* (answer directly on short inputs, decompose on long ones): absolute and relative cost, KL to the warm start, sequence normalisation, short-task weighting, stratified sampling, a balance term.

The decisive pair

A consistent teacher makes the rule learnable by supervised fine-tuning (direct on 38 of 40 short tasks, decompose on 31 of 32 long ones). RL started *from that correct router* is all-direct by step 5 (dev 0.87 to 0.30).

The rule is representable; policy-gradient dynamics move it globally. Routing collapse is also reported in the literature (Kassenaar et al. 2026).

Ledger E06, E07-1 to E07-9 in docs/RESULTS.md. One seed per arm except where stated; synthetic families; exact-match scoring.

Put the routine in a library: the untrained controller matches the trained ones.

The kernel now runs a *prelude*: library code bound before the first cell, listed in the prompt by name and contract, never shown as source. The controller only has to call it.

Store log, dev split (96 aggregation and 24 lookup tasks)	Aggregation	Lookups	Tokens/task
Scripted expert (full programs in its cells)	88/96	24/24	17.0k
Taught controller SFT-005 (routine in the weights)	86/96	24/24	15.8k
Untrained 4B with no library	46/192	36/48	—
Untrained 4B + prebound library	**85/96**	**23/24**	16.8k
Controller taught to call the library (SFT-009)	85/96	24/24	14.9k

OOLONG-synth, 293-row held-out set	Accuracy	Errors	Tokens
Direct answer / taught SFT-005	0.208 / 0.567	0 / 31	12.9k / 10.3k
Best RL controller (E06)	0.652	5	14.3k
Untrained 4B + OOLONG library (expert: 0.672)	**0.672**	**0**	**11.0k**

Teaching the call pattern (SFT-009) adds only the cheap lookup route, at four times the tokens on short tasks.

Ledger L1-b to L1-d, L4, E09. No-library row: baseline sweep (train split). OOLONG: 293 held-out rows, six lengths from 1k to 32k tokens.

Put the route choice in a table: no training, best accuracy, lowest cost.

The table. Every logged run on the teaching split contributes (input signature, routine, correct, tokens). The signature is computed from the input text alone: family, line-count bucket, question form.

The note. At run time the harness retrieves the history for the input's signature and renders it in the prompt with one recommendation: highest accuracy minus 0.05 per 10k tokens.

Result. The untrained controller follows the recommendation on 120 of 120 dev tasks.

Dev split	Agg.	Lookups	Tokens
Untrained + library	85/96	23/24	16.8k
lookups alone			17.3k
+ statistics note	**87/96**	**24/24**	**14.3k**
lookups alone			**1.6k**
Scripted expert	88/96	24/24	17.0k

Online, and with a learned key

Online streams: greedy warm-started from the table **0.87**, lowest regret. Learned key (the controller's own hidden state + length, nearest neighbours) matches the hand signature on 413 held-out tasks.

Ledger L3a; skills/stats.py; table from teaching and train splits only; online streams: runs e10-bandit-*.

A library must not overclaim: what happens off its coverage.

Three question forms the library does not cover were added to the generator (most-buying person, purchase span, busiest day with a tie rule).

72 held-out dev tasks	Correct	What the controller does
Direct answer, no library	29/72	fine to 64 lines, near zero above
Untrained + shipped library (docstring claims every form)	0/72	submits the library's "unknown"
Untrained + honest library (None off coverage; look first)	34/72	writes its own aggregation
of which top buyer	1/24	right name, wrong format in 20 of 24

Reading. A trusting controller plus an overclaiming library is worse than no library. An honest contract recovers most of it; the residual losses, exact output format and budget-safe chunking, are what a verified routine provides.

Ledger E08-0, E08-1. A first honest variant let the controller guess before it had seen the question (0/3); the fix: look first, aggregate next cell.

Grow the library by verification: the librarian loop.

1. Cache the model's own event extractions for the teaching tasks of the new form.
2. Ask the proposer for candidate routines: event schema, the shipped aggregator as style, three teaching examples with gold answers.
3. Verify each candidate on the cached extractions against gold; keep the best above threshold.
4. Register it with its verification record; it enters the statistics table like any route.

Result: four held-out forms

Untrained caller on dev, before → after growth:
top buyer 0 → **24**/24; purchase span 0 → **24**/24 (4B proposer);
busiest day 0 → **24**/24: every 4B candidate had one sort bug (0/48, kept out); an 8B proposer fixed it;
join (bought on a day someone else returned) 0 → **22**/24 (4B, 43/48 verified; direct answer 6/24).

The ceiling is the proposer, not the caller; verification makes mixing capacities safe.

Runs e08-*, e11-*. Prior art: Voyager, LILO, Agent Workflow Memory.

How small can the caller be? Down to 0.6B, if it learns the call.

Helpers fixed at the 4B; library and statistics note throughout. Correct answers:

Caller	Dev (120)	Grown forms (96)	OOLONG-72
0.6B / 1.7B untrained	11 / 77	18 / 28	1 / 1
4B untrained	111	94	43
8B (4-bit) untrained	112	93	43
Phi-4-mini (3.8B) untrained	69	94	0
0.6B tuned on the 4B's one-cell calls (two seeds)	**110 / 107**	**94 / 93**	**43 / 42**
Phi-4-mini tuned, same 436 traces	105	94	43
SmolLM2-360M tuned, same traces	109	48	42

On all 293 held-out OOLONG rows the tuned 0.6B and 1.7B and the untrained 4B are right on *identical* rows (0.672). 55 traces suffice; they must include the family. At 360M the caller misreads the note's routine on two of four new forms.

Ledger S1 to S5. First fine-tunes voided by a chat-template mismatch.

The caller never reads the input: from 1k to 2M tokens.

OOLONG-synth, 0.6B caller	Correct	Tokens/row
64k (20 rows)	15	65k
128k (20)	16	134k
256k (20)	12	265k
512k (20)	7	515k
1M (8)	5	1.0M
2M (4)	2	2.0M

Cost is linear in length. Store logs of 20k to 80k tokens: library 98/108, direct 3/108. OOLONG declines through count questions: the helper mislabels about 1 row in 10 (spam) to 1 in 5 (TREC).

The RLM paper's setting

OOLONG trec_coarse, 131K tokens, 50 tasks:

0.6B caller + library: **45.3** (exact 44.0)
GPT-5 base (paper): 44.0
RLM(GPT-5) (paper): 56.0 to 58.0

Released RLM-Qwen3-8B in its own scaffold, our 293 held-out rows: 0.290, about 0.32 with a 131k window (library: 0.672; direct: 0.208).

No routine variant, larger helper, correction or helper tuning lifted the count ceiling.

Ledger E14 to E20. Paper numbers from its Table 1; its scoring rule is not stated (ours: OOLONG partial credit).

What this changes, and what to test next.

The claim, sharpened. For a small controller, capability belongs in a library of verified programs over the kernel's ABI, selection in outcome statistics keyed by the input; the weights only call and compose. RL's remaining job is where verification is unavailable: helper quality, and proposing routines the librarian then verifies.

Limits. Synthetic families; one seed per arm; exact-match scoring; a greedy table keyed by hand-chosen signature features; growth tested on easy forms first.

Question	Smallest test
Harder growth?	multi-step forms and a second family; proposer size vs acceptance
Learned signature?	router over the controller's own state instead of hand-chosen features
External validity?	OOLONG-real with the library; the 293-row held-out run
Confirm the 0.6B?	293-row OOLONG; second seed; other model families

Plan of record: docs/research-plans/2026-10-20-revised-plan-and-prior-art.md. Allocation ends 22 October 00:39 UTC.

Appendix: protocol and provenance.

Item	Protocol
Models	Qwen3-4B-Instruct-2507 as controller and helper, served by vLLM 0.27.1 with LoRA hot-loading; two A100 40GB.
Families and splits	Store log: train, teach (seeds 16 to 23) and dev (seeds 1000 to 1003) by seed namespace; OOLONG-synth: even context windows teach, odd held out; 72-row sample across six lengths.
Library	Event-log skills: split, chunked extraction by helper calls, exact aggregation, cheap lookup; OOLONG skills: parse, classify, aggregate. Bound by the kernel prelude, digest in the harness fingerprint, hidden from the variable listing.
Statistics	Routine classified from the controller's cells in the trace; costs are total tokens; scripted replays get a 1.5k controller overhead added.
Scoring	Exact match after normalisation (integers; sorted name lists). Failed and aborted runs preserved with notes.

Code: github.com/queelius/rlm (experiments/selective_recursion_20261019, src/rlm); ledger docs/RESULTS.md; notebook docs/notebook/2026-10-19-*.md.